

EMEC -2



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Experiment No. 6

Speed control of 3-phase Induction motor by cascading.

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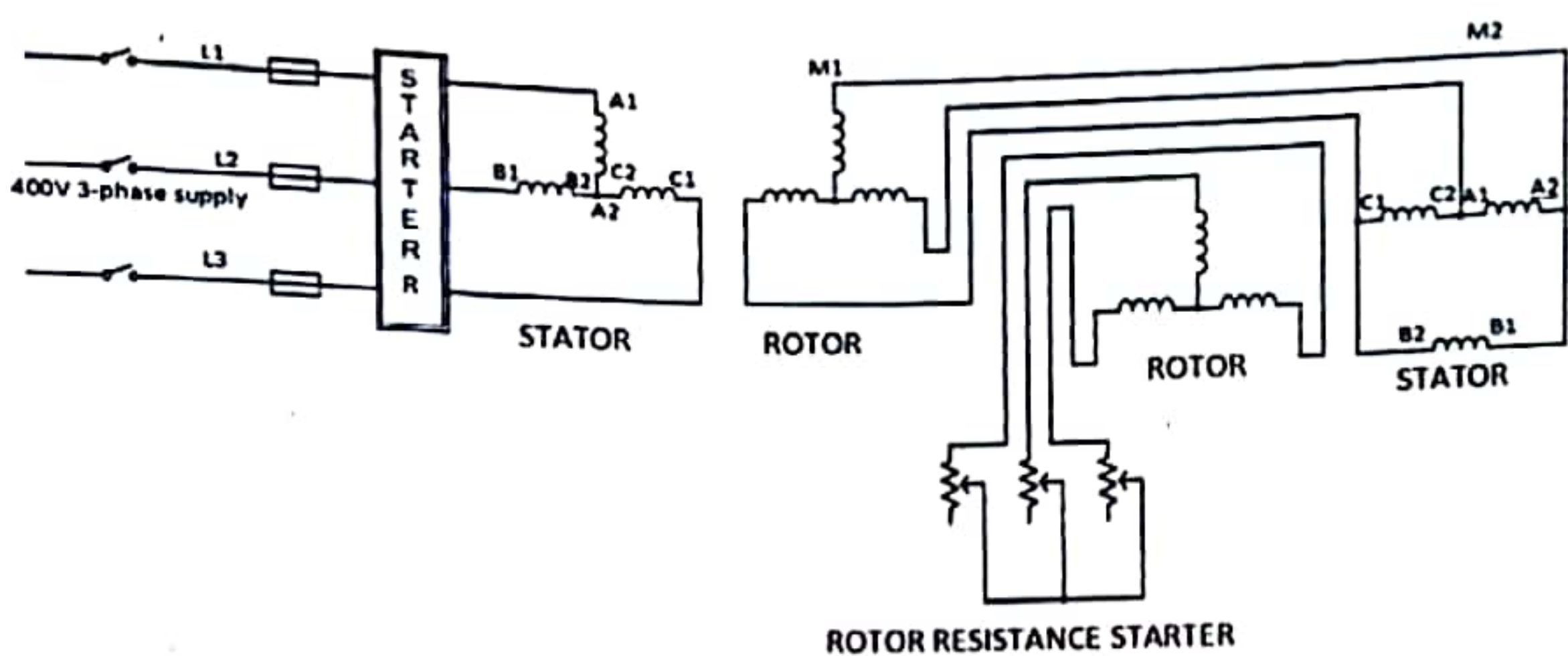
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Experiment No- 6

Aim:

Speed control of 3-Ø Induction motor by cascading.

Circuit diagram:



Theory:

An induction motor is basically a constant speed motor over the load range where two induction motors are available (one of them must be slip ring type), they are cascaded together to obtain four different speeds which remain fairly constant over the load range. If P_1 and P_2 are the number of poles of the two motors, the four different speeds are;

$$N_1 = 120f/P_1$$

$$N_2 = 120f/P_2$$

$$N_3 = 120f/P_1 + P_2$$

$$N_4 = 120f/P_1 - P_2 \quad \{\text{where } P_1 \neq P_2\}$$

In cascading, the rotor of one of the machines (slip-ring) is connected to the stator of the other. Both motors mechanically coupled and the second motor will open at the slip frequency and rotor voltage of the first.

The motors are said to be cumulatively copied if the rotating magnetic field and torque are both in same direction. This gives speed N_3 , if the rotating magnetic field in the two motors are in opposite direction, they are said to be differentially coupled. This gives the speed N_4 , which is usually very high and is not generally preferred in view of the low torque output.

Procedure:

1. Connections are made as shown in figure.

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2. Start M with the help of a starter after ensuring that a large value of resistance is included in the rotor circuit of M2 (this is applied only if M2 is of the slip ring type).
3. This will help to limit the current during starting. The rotor resistance is cut in steps and increase in speed is noted.
4. The phase sequence of the input to the second motor is reversed and the experiment is repeated.
5. The machines are then started individually and the speeds are noted. The readings are tabulated as shown in table.

Observation table:

Motor no.	Run As	Theoretical speed	Measured speed
M1	Individual		
M2	Individual		
M3	Cumulative Cascade		
M4	Differential Cascade		

Conclusion:

1. In differential cascade method, we get maximum speed.
2. In cumulative cascade, we get minimum speed.

Precautions:

- 7) Connections must be done as per circuit diagram.
- 8) Correct range of instruments must be connected.
- 9) Voltage applied through secondary of transformer must be kept low in the starting.
- 10) Connection of current coil of wattmeter must be reversed if reading in negative.
- 11) Current must not exceed the rated current.
- 12) Voltage must be kept constant during the load test.